

EX.NO:	Application Using Layout Managers & Event Listeners
DATE: / /2025	

AIM:

To develop a simple Android application using **layout managers** to organize UI components and **event listeners** to respond to user actions.

ALGORITHM:

Step-1: Open Android Studio and make a new project. Choose Empty Activity.

Step-2: Give the project a name (for example: LayoutApp) and finish creating it.

Step-3: In the project, open the layout file: app/src/main/res/layout/activity_main.xml.

Step-4: In that XML file, put a LinearLayout (vertical). Inside it add:

- A TextView (shows some text). Give it id @+id/textView.
- A Button (you can press it). Give it id @+id/buttonClick.

Step-5: Save the layout file. It will look like a vertical column with text on top and a button below.

Step-6: Now open the activity code file: app/src/main/java/.../MainActivity.java (or .kt for Kotlin).

Step-7: Inside onCreate() make the activity use the layout:

- setContentView(R.layout.activity_main);

Step-8: Get the TextView and Button from the screen by their ids:

- TextView textView = findViewById(R.id.textView);
- Button button = findViewById(R.id.buttonClick);

Step-9: Tell the button to listen for clicks:

- button.setOnClickListener(...)

Step-10: Inside the click code, tell the TextView to change its text:

- textView.setText("Button Clicked!");

Step-11: Save your code.

Step-12: Run the app on an emulator or a real phone.

Step-13: When the app opens you will see the text and the button. Tap the button — the text will change.

SOURCE CODE:

activity_main.xml:

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:padding="20dp">

    <TextView
        android:id="@+id/textView"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Click the Button"
        android:textSize="24sp"
        android:textColor="#000000"
        android:padding="10dp" />

    <Button
        android:id="@+id/buttonClick"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Click Me"
        android:padding="10dp" />

</LinearLayout>
```

MainActivity.Java:

```
package com.example.layoutapp;

import androidx.appcompat.app.AppCompatActivity;
import android.os.Bundle;
import android.view.View;
import android.widget.Button;
import android.widget.TextView;

public class MainActivity extends AppCompatActivity {

    TextView textView;
    Button buttonClick;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        textView = findViewById(R.id.textview);
        buttonClick = findViewById(R.id.buttonClick);

        // Event Listener (Button Click)
        buttonClick.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View v) {
                textView.setText("Button Clicked!");
            }
        });
    }
}
```

AndroidManifest.xml:

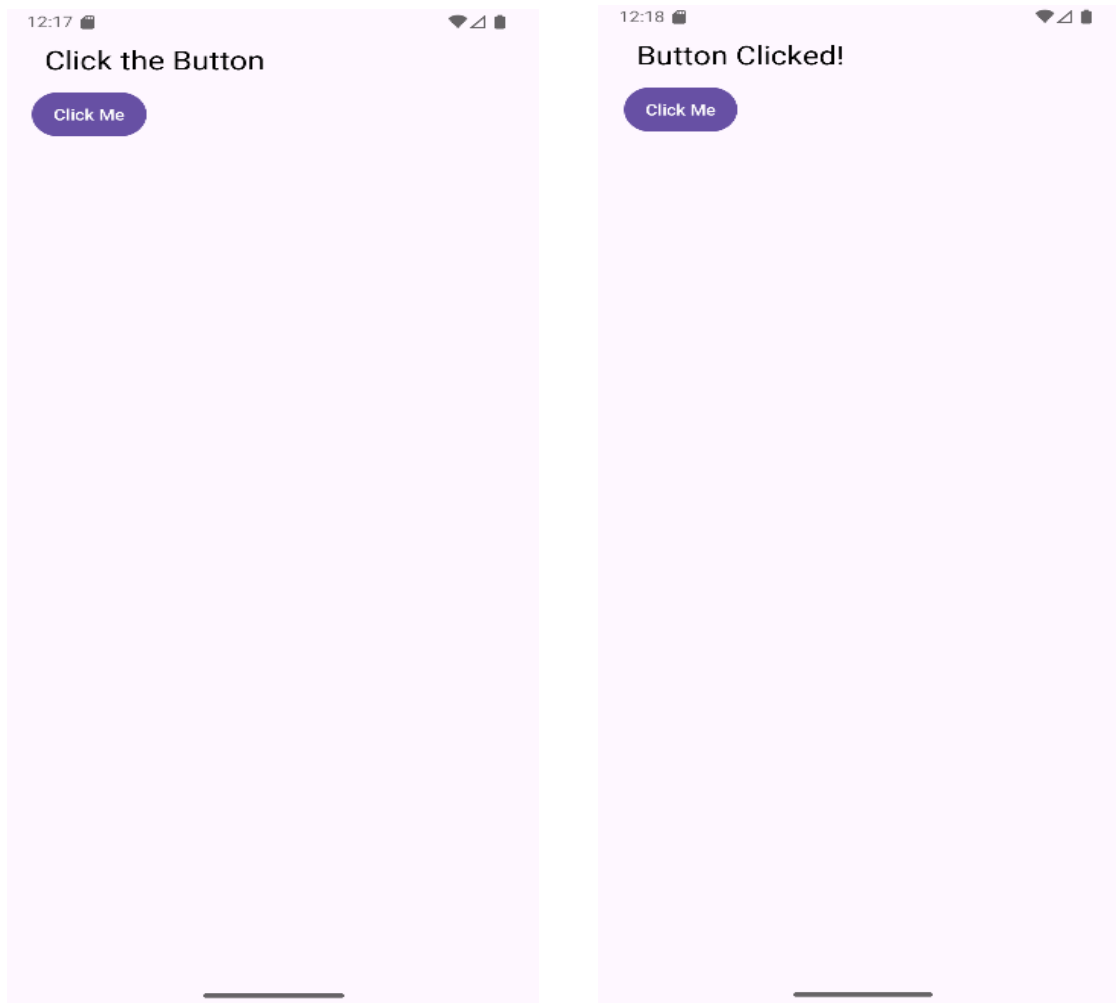
```
<?xml version="1.0" encoding="utf-8"?>
<manifest xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:tools="http://schemas.android.com/tools">

    <application
        android:allowBackup="true"
        android:dataExtractionRules="@xml/data_extraction_rules"
        android:fullBackupContent="@xml/backup_rules"
        android:icon="@mipmap/ic_launcher"
        android:label="@string/app_name"
        android:roundIcon="@mipmap/ic_launcher_round"
        android:supportRtl="true"
        android:theme="@style/Theme.Layoutapp">
        <activity
            android:name=".MainActivity"
            android:exported="true"
            android:theme="@style/Theme.Layoutapp">
            <intent-filter>
                <action android:name="android.intent.action.MAIN" />

                <category android:name="android.intent.category.LAUNCHER" />
            </intent-filter>
        </activity>
    </application>

</manifest>
```

OUTPUT:



RESULT:

The Android application was successfully developed using **Layout Managers** and **Event Listeners**.